

OpenBCI Recording: Data Quality Assessment

Why this recording cannot be used to train a muscle-pattern model yet

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Bottom line

We did a signal-level analysis of the recording before training. There is a data-quality problem that prevents meaningful training on this file. This is not a modelling failure and not a dead end: it is fixable with one clean re-recording plus a small preprocessing change. The pipeline we have already built will then become meaningful.

Finding 1: the amplifier was railed (saturated) for 70% of the session

In 70.5% of all samples, at least one of the eight channels is pegged at the hardware rail (the Cyton's full-scale limit, plus or minus 187500 microvolts at gain 24). A railed input is not signal: it is a clipped, flat-topped reading, caused by electrode contact, impedance, or gain. As the figure shows, the input sits at the rail for roughly the first 390 seconds and only settles into a sane range in the final ~150 seconds.

A side effect: clipping is a square wave, so it dumps energy into every frequency band. The high-frequency "bursts" that could look like muscle activity are clipping artifact, not EMG.

Finding 2: where the data is clean, the real muscle signal is weak and unfiltered

In the ~30% of the recording that is not railed, there is some genuine bicep EMG, but it is weak and buried under low-frequency baseline drift. Even in the best clean windows the muscle-band (20 to 100 Hz) power never rises above about 35% of the sub-5 Hz drift power, and the current pipeline applies no high-pass filter to remove that drift. There was also no timed flex/rest protocol, so the muscle activity has no clear structure to learn from. In short, on this recording we cannot reliably tell muscle activity from drift and noise.

Finding 3: the model scores do not survive a fair test

We also checked the models themselves on this data:

- The forecasting LSTM reports RMSE 0.1945. A one-line linear extrapolation ("continue the straight line") scores 0.1696 on the same test segment, which is better. The LSTM is not beating a trivial baseline.
- The classifier reports 97.6% accuracy. That figure comes from overlapping windows split randomly between training and test, which leaks near-duplicate data across the split. Under an honest split (hold out whole segments), a trivial three-feature logistic regression already reaches about 80%, and the LSTM adds nothing on top of the correlation that defines the labels.

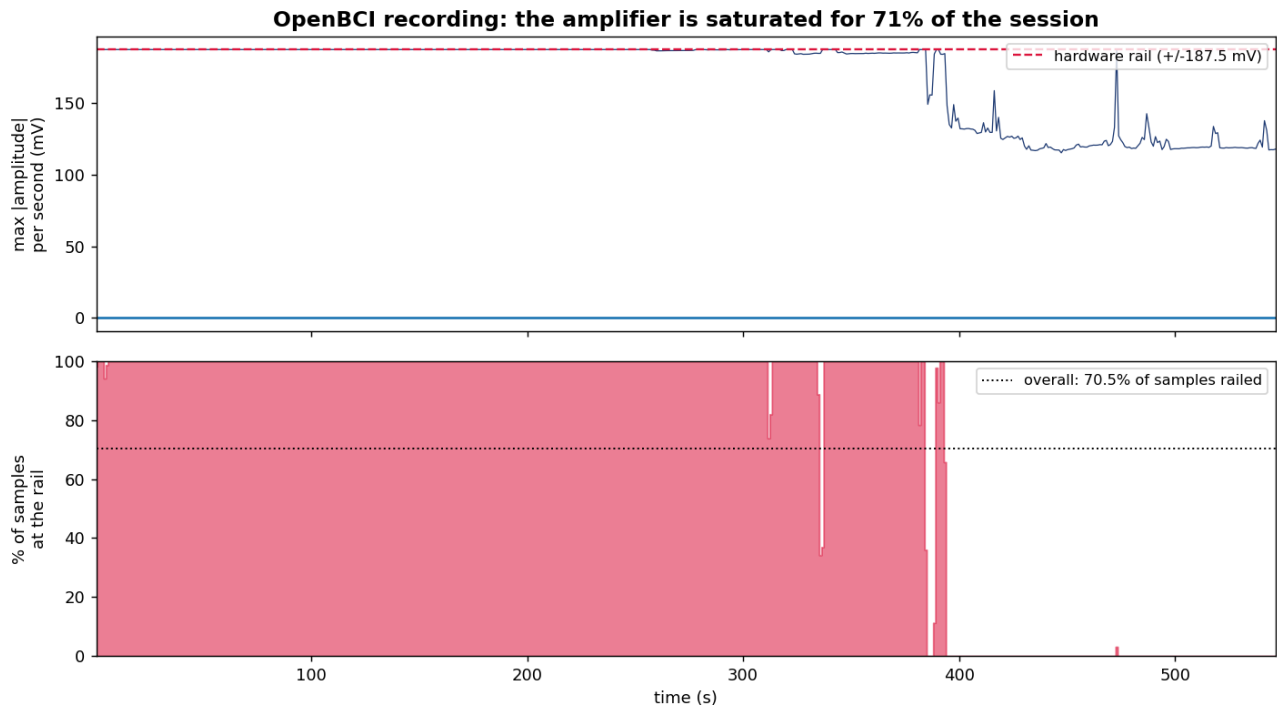


Figure 1: Amplitude over the recording. The signal sits at the hardware rail (red dashed line) for most of the session; the lower panel shows the percentage of samples railed per second.

So the strong numbers reflect the test setup, not learned muscle patterns.

What we recommend

The rig and the pipeline are fine. The recording is the problem. To get trainable data we need:

1. **Fix the railing at the source** (electrode contact, skin preparation, impedance, gain) so channels sit in a sane microvolt range and stop clipping.
2. **Add a high-pass filter** (around 10 to 20 Hz) in preprocessing to remove the baseline drift that currently swamps the muscle signal. This is a code change.
3. **Record with a timed protocol** (for example 5 seconds rest, 5 seconds flex, repeated) so the muscle bursts are high-contrast and labellable, and define a “valid pattern” by muscle-band activity during flex rather than by raw channel correlation.

With those three changes, one good recording gives us data we can actually train on, and the existing scripts (now corrected for the leakage and with proper baselines) will produce numbers that mean something.

Supporting analysis, plots, and the corrected scripts are available on request.